

FEATURED PROJECTS

Formula SAE - UP Racing

Student group focused on developing a formula-style single-seater vehicle for FSAE competition in Michigan.

Pedal System (accelerator, brake, clutch)

Complete pedal system design for a Formula-style competition car (FSAE). Comprehensive research and justification, including hand calculations and programs for automation, design, and simulations.

Recumbent

Overall project leader. Design of a recumbent-style tricycle, including fatigue and natural frequency analysis.

RC Vehicle with Launcher

Development of an RC vehicle with a fully mechanical launcher, servo-motor controlled, with an integrated camera for remote operation, and tank tracks for greater versatility.

EDUCATION

Bachelor's Degree in Mechatronics Engineering

Universidad Panamericana - Guadalajara
2023 - Present
GPA: 9.62

SKILLS

Analysis and simulation: ANSYS, PrePoMax

CAD: SolidWorks, OnShape

Programming: C++, MATLAB, Octave

Tools: Microsoft Office, 3D Printing

Technical interests: Engineering design, data analysis, process optimization, and prototyping.

Soft skills: Teamwork, communication, problem-solving, leadership, time management, and adaptability.

LANGUAGES

Spanish — Native

English — Advanced (C1),
valid TOEFL ITP score of 643

PROFESSIONAL PROFILE

20-year-old Mechatronics Engineering student, passionate about technology, innovation, and the development of solutions applied to mechanical and automotive systems. I have experience in prototyping, computational tools, and multidisciplinary projects, participating in both technical development and team coordination. I am characterized by initiative, self-directed learning, analytical thinking, and a problem-solving mindset.

EXPERIENCE

Vehicle Dynamics Area Leader

UP Racing - Formula SAE Team
August 2024 - 2025

- Training and onboarding of a new team.
- Development of MATLAB and Octave tools to streamline engineering calculations and analysis.
- Analysis and validation of suspension, steering, and braking systems, including the implementation and design of the braking system.
- Research, justification, design, and simulation of a pedal system (brake, accelerator, and clutch).

Mechatronics Class Representative

August 2025 - Present

- Representation of the class before university directors and academic coordinators.
- Organization of activities and follow-up on student initiatives through the committee.

Tech Fest Organizer

Presentation of mechatronics projects
July 2026

- Leadership of the organizing committee for the first edition of Tech Fest.
- Coordination of logistics, spaces, and communication with security and support teams.

University Academic Advisor

August 2025 - Present

- Personalized tutoring in Statics, Dynamics, Electrical Circuits, and Programming.
- Support in developing analytical and problem-solving skills.

CERTIFICATIONS & ACHIEVEMENTS

Certified SOLIDWORKS Associate (CSWA)

Copilot: Master AI in Microsoft 365

1st place in the UP Parabolic Shooting Competition - 50% scholarship
3x Top Highest GPA in the Mechatronics class